

PORTSMOUTH PEASE, NH, USA

WMO: 726055

Lat: 43.083N

Lon: 70.817W

Elev: 100

StdP: 14.64

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 04743

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.0	7.3	-11.2	3.0	7.2	-6.9	3.8	11.1	26.9	25.1	24.9	22.9	11.0	290	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.1	89.6	72.7	86.2	71.3	83.2	69.8	75.3	85.2	73.6	82.1	72.2	79.8	9.6	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	120.4	79.7	71.2	115.3	78.3	69.8	109.9	76.7	38.8	85.4	37.3	82.1	35.9	79.6	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.5	20.1	17.8	DB	-4.5	94.4	4.9	2.5	-8.0	96.2	-10.9	97.7	-13.6	99.1	-17.2	100.9	(4)
(5)			WB	-5.5	78.5	4.7	1.4	-8.8	79.5	-11.5	80.3	-14.2	81.1	-17.6	82.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.1	25.2	28.3	35.4	46.5	56.2	65.7	71.8	70.6	63.2	51.8	41.3	31.9		
	DBStd	17.79	10.39	9.50	8.96	7.74	7.26	7.00	5.37	5.13	6.81	7.22	8.22	8.83		
	HDD50	2924	770	607	465	156	17	0	0	0	1	64	281	563		
	HDD65	6390	1235	1027	920	559	293	75	7	10	113	412	712	1027		
	CDD50	2592	0	1	11	49	209	471	675	638	396	120	20	2		
	CDD65	580	0	0	1	3	19	95	217	183	58	4	0	0		
	CDH74	4625	0	0	14	55	248	820	1719	1332	405	32	0	0		
	CDH80	1415	0	0	4	15	78	249	570	390	107	2	0	0		
(14) Wind	WSAvg	8.1	9.3	9.3	9.6	9.3	8.1	7.1	6.7	6.5	6.7	7.8	8.2	8.6		
(15) Precipitation	PrecAvg	46.1	3.3	3.2	4.2	4.0	3.7	4.0	3.7	3.5	3.9	4.7	3.9	4.1		
	PrecMax	65.9	6.5	8.4	15.3	10.2	15.2	12.4	8.6	10.6	10.4	14.1	7.6	6.4		
	PrecMin	33.7	0.6	1.3	0.8	0.4	0.7	1.1	0.9	0.2	1.3	0.3	1.2	1.3		
	PrecStd	8.5	1.1	1.5	2.8	2.2	2.8	2.5	2.0	2.0	2.2	3.3	1.5	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.3	57.5	72.7	81.4	87.8	91.1	93.2	91.8	89.0	78.6	68.4	60.1		
		MCWB	53.5	51.0	57.6	61.2	68.0	73.6	75.3	74.3	72.2	64.8	59.9	52.2		
	2%	DB	48.3	50.3	60.5	71.9	81.2	86.5	89.6	87.8	82.4	72.3	62.8	53.0		
		MCWB	44.9	43.6	51.1	57.4	64.2	71.3	73.4	73.0	69.6	62.5	55.8	48.8		
	5%	DB	43.3	45.6	52.4	64.7	74.9	82.2	86.2	84.2	78.7	67.6	58.4	48.8		
		MCWB	39.3	39.9	44.5	52.9	61.3	68.5	71.8	70.7	67.1	59.7	52.2	45.3		
	10%	DB	38.9	41.2	47.4	59.4	69.7	78.8	82.8	81.4	74.6	63.4	54.2	44.7		
		MCWB	35.3	36.4	40.9	49.0	58.6	66.3	70.2	69.3	65.8	56.6	49.3	40.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	53.0	59.1	64.2	70.8	76.2	78.0	76.7	74.8	68.7	61.8	54.8		
		MCDB	56.3	55.4	70.9	75.0	83.4	86.5	89.0	87.3	83.7	73.0	66.1	58.0		
	2%	WB	45.9	45.5	52.3	59.9	67.1	73.7	75.7	74.7	72.2	64.4	57.8	50.2		
		MCDB	47.2	49.2	59.3	70.0	76.6	83.4	85.8	83.6	79.3	69.3	60.8	52.0		
	5%	WB	40.1	40.5	45.7	54.9	63.4	71.4	73.9	73.1	70.1	61.2	53.4	45.8		
		MCDB	42.8	44.4	50.5	62.9	71.8	78.9	82.5	80.7	75.6	65.9	56.5	47.9		
	10%	WB	35.5	37.2	41.6	50.3	59.9	69.0	72.2	71.4	67.9	58.3	49.9	41.1		
		MCDB	38.2	40.6	46.6	57.7	67.7	75.4	79.8	78.6	72.7	62.3	53.6	44.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	16.1	15.8	18.2	18.2	18.2	18.1	18.3	18.1	17.3	15.5	14.3		
		MCDBR	18.6	20.3	23.2	27.9	27.5	23.4	22.2	22.0	22.5	23.0	20.6	19.2		
	5% WB	MCWBR	16.8	15.1	15.3	16.1	14.0	11.3	9.6	9.7	11.4	14.1	15.3	16.1		
		MCWBR	18.5	18.9	22.0	24.8	23.9	21.4	19.4	18.5	18.8	19.2	17.6	18.7		
(40) Clear-Sky Solar Irradiance	taub	taub	0.286	0.296	0.319	0.357	0.394	0.441	0.459	0.426	0.374	0.332	0.311	0.291		
		taud	2.487	2.477	2.486	2.423	2.359	2.260	2.215	2.295	2.424	2.540	2.558	2.550		
	Ebn at Noon	Ebn at Noon	273	289	294	289	279	265	259	264	271	272	262	262		
		Edh at Noon	23	27	30	35	38	43	44	39	32	25	21	20		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	541	817	1154	1464	1633	1758	1851	1677	1348	876	584	438		
	RadStd	RadStd	36	64	75	158	184	155	120	79	82	78	51	48		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	-2.74	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+149	+59

Nomenclature: See separate page